

UPSERT

UPSERT is asked very frequently in backend and database interviews because it's used in almost every production application.

What is UPSERT?

UPSERT = UPDATE + INSERT

It means:

- If the row doesn't exist → **INSERT** it.
- If the row already exists → **UPDATE** it.

Instead of writing:

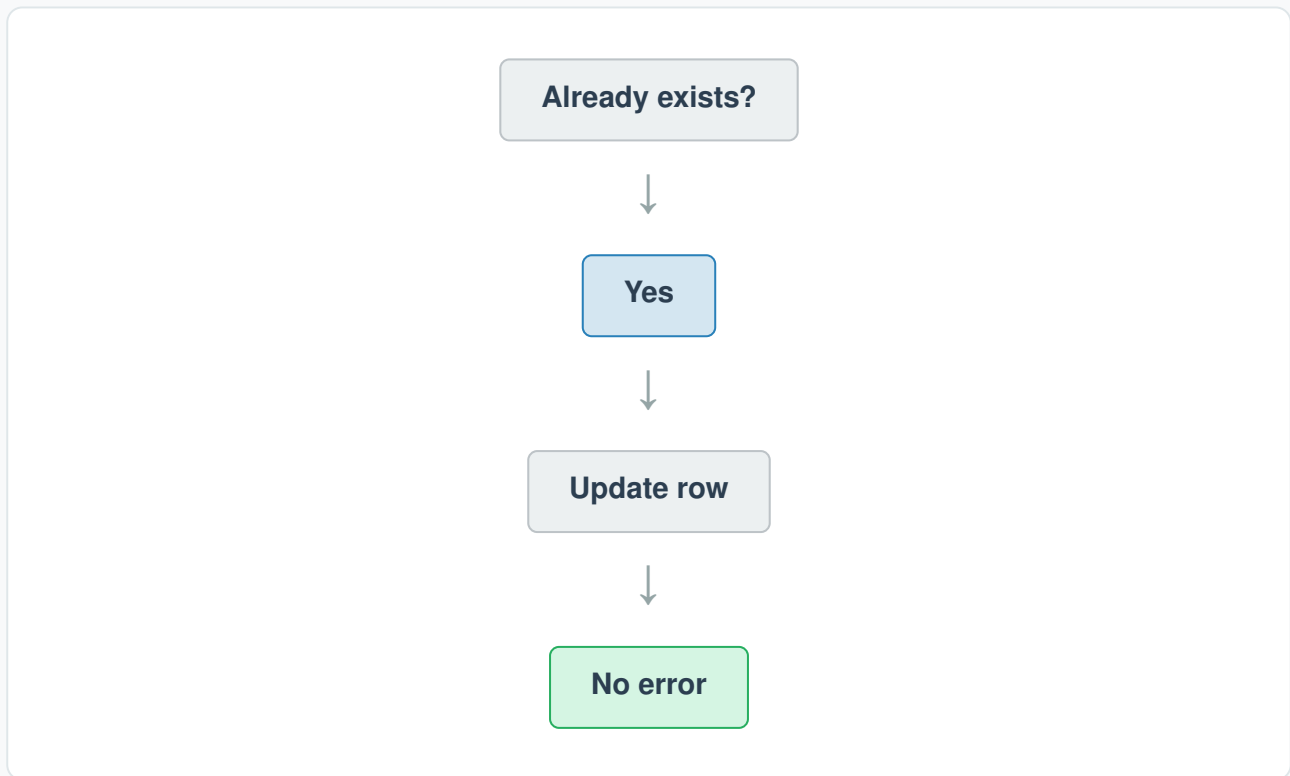
```
IF EXISTS
    UPDATE
ELSE
    INSERT
```

SQL provides a single statement.

Suppose

Students	
StudentID	Name
1	Aadya

UPSERT instead says



INSERT ... ON CONFLICT (PostgreSQL)

```
INSERT INTO Students
VALUES (1, 'Arpit')
ON CONFLICT(ID)
DO UPDATE
SET Name='Arpit'
```

ON DUPLICATE KEY UPDATE (MySQL)

```
INSERT INTO Students(StudentID, Name)
VALUES (1, 'Arpit')
ON DUPLICATE KEY UPDATE
Name='Arpit';

INSERT INTO Students(StudentID, Name)
VALUES (1, 'Arpit')
ON DUPLICATE KEY UPDATE
Name='Arpit';
```

Updating Using the Inserted Values

Suppose

Products	
ID	Stock

- **Existing**
Stock = 10
- **Shipment**
15
- **Increase stock**

```
INSERT INTO Products(ProductID,Stock)
VALUES (1,15)
ON DUPLICATE KEY UPDATE
Stock = Stock + VALUES(Stock);
```

DO NOTHING

Sometimes
You don't want update.
You simply want

Ignore duplicate

PostgreSQL

```
INSERT INTO Students
VALUES(1, 'Aadya')
ON CONFLICT(ID)
DO NOTHING;
```

Equivalent MySQL idea

```
INSERT IGNORE INTO Students
VALUES(1, 'Aadya');
```

DO UPDATE

Instead of ignoring
Update

PostgreSQL

```
ON CONFLICT(ID)
DO UPDATE
SET Name='Aadya' ;
```

MySQL

```
ON DUPLICATE KEY UPDATE
Name='Aadya' ;
```

When Conflicts Happen

UPSERT only works when a conflict can occur.
Conflicts happen on

- **Primary Key**
StudentID
- **Unique Key**
Email
- **Composite Key**
Example (StudentID, CourseID)

```
INSERT INTO Users(UserID,LoginCount)
VALUES(10,1)
ON DUPLICATE KEY UPDATE
LoginCount=LoginCount+1;
```

UPSERT Multiple Rows

```
INSERT INTO Students
VALUES
(1, 'Aadya'),
(2, 'Arpit'),
(3, 'aadz')
ON DUPLICATE KEY UPDATE
Name=VALUES(Name);
```

Each row is checked independently.

Some may insert.

Some may update.

Real-World Examples

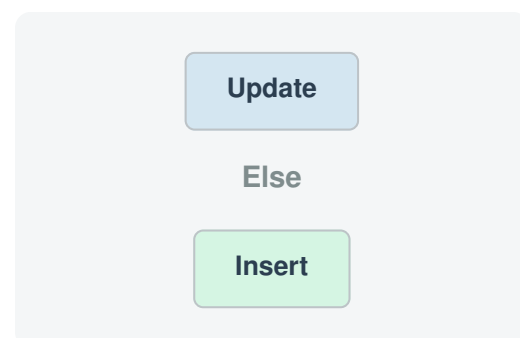
Amazon

Shopping cart



WhatsApp

Contact already exists?



Summary

Why use UPSERT instead of SELECT + INSERT/UPDATE?

- Fewer queries
- Atomic
- Better performance

MySQL syntax?

```
INSERT INTO table(...)
VALUES(...)
ON DUPLICATE KEY UPDATE
column = value;
```

4. PostgreSQL syntax?

```
INSERT ...
ON CONFLICT (...)
DO UPDATE;
```

SQL Server equivalent?

MERGE